

Exploring a Shared Augmented Reality Experience for Co-located Co-Design and Maintenance on the Shop Floor

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ABSTRACT

Industrial maintenance plays a critical role for operational efficiency, prolonging equipment lifespan, and minimizing unplanned downtime that can disrupt production workflows. Incorporating co-design practices into this domain is equally important, as it facilitates the development of tools and processes that are better aligned with how maintenance procedures are actually carried out. Augmented Reality (AR) technologies have proven effective in supporting such scenarios, enhancing training, and reducing downtime. Nevertheless, in such contexts, certain tasks require a coordinated team effort, fostering a collaborative approach where multiple technicians work together to diagnose issues, share insights, and implement solutions efficiently. Yet, most AR studies focus on individual assistance, validated under controlled laboratory settings with simplified scenarios that do not accurately represent real-world challenges. This work proposes a collaborative AR tool that enables various co-located technicians to interact synchronously. The tool facilitates shared 3D object manipulation, decomposition of complex models, precise measurements, and sketching annotations for effective communication. These features establish a common ground among team members, ensuring a unified understanding of the task at hand. As a use case, a heavy hydraulic press was considered, a crucial component of the production line, selected following a Human-Centered Design (HCD) methodology in collaboration with industry partners. The tool was validated through iterative testing with maintenance technicians and Industry 4.0 engineers, ensuring alignment with real-world needs and expectations.

Keywords: Augmented Reality, Co-Design, Maintenance, Shop Floor, Industry 4.0, Collaboration, Shared Experience

1 INTRODUCTION

Maintenance is paramount in industrial scenarios, defined as the set of activities required to ensure that machinery and equipment remain

in optimal operating conditions [12, 20]. It is essential not only for preventing unexpected breakdowns but also for extending equipment lifespan and ensuring consistent production output [1, 3, 14]. With the increasing complexity of industrial environments brought on by the rise of Industry 4.0, maintenance processes have become even more critical to avoid costly downtime and disruptions [8, 9, 19].

Equally important in supporting maintenance efforts are co-design activities, which are fundamental for tailoring procedures to the real-world needs of technicians [17, 21]. By involving end-users in the design and refinement of tools and workflows, co-design helps ensure that maintenance processes are more intuitive, efficient, and aligned with the specific challenges encountered on the shop floor [2, 11].

One technology that holds significant promise for such scenarios is Augmented Reality (AR), which is considered a key enabler of the latest industrial revolution [15, 16, 18]. This technology supports the overlay of digital content onto physical environments, allowing to visualize additional information, receive contextual guidance, and interact with virtual content in real time. These capabilities can lead to benefits such as enhanced training for maintenance tasks, faster and more accurate diagnostics, improved collaboration during co-design sessions, reduced errors, increased efficiency, and minimized downtime in industrial operations [5, 6, 10].

However, most AR studies have primarily focused on individual assistance in maintenance scenarios, overlooking collaborative settings, while co-design activities are typically explored within Virtual Reality (VR) environments, detaching users from the real-world context and requiring them to be off-site. Furthermore, the vast majority of these works have been conducted in controlled laboratory settings using simplified scenarios that fail to reflect the complexity of real shop floor conditions [4, 7, 13, 16]. This highlights a clear opportunity to investigate collaborative scenarios directly within industrial environments, where several practical challenges remain.

This work proposes a tool to support collaborative scenarios using shared Augmented Reality. It allows multiple co-located members to interact synchronously, manipulate 3D objects, decompose complex models, perform precise measurements, and add annotations. These help establish a common ground, ensuring a shared understanding of the problem at hand. A heavy hydraulic press (essential for a the production line) was selected as the primary use case, identified through a Human-Centered Design (HCD) approach in collaboration with industry partners. The tool was validated with domain experts to ensure it meets real-world needs and expectations.

2 COLLABORATIVE AR TOOL FOR SHARED EXPERIENCES

To explore the use of collaborative AR for co-design and maintenance, a use case centered on a heavy hydraulic press was selected, following a HCD methodology with industry partners (Figure 1). The hydraulic press plays a pivotal role in shaping metal sheets, producing components that are essential for the factory's final commercial products. This is a large, heavy, and complex machine,

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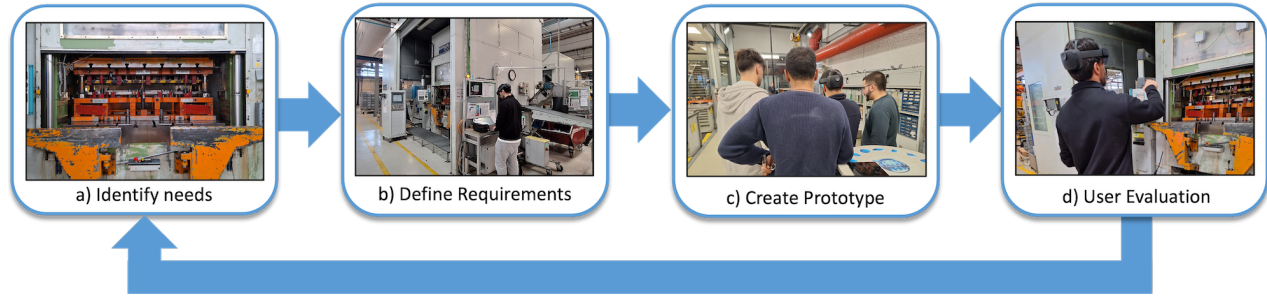


Figure 1: Methodology adopted to bring domain experts into the design of a collaborative AR tool: a) identify needs of target users; b) definition of requirements; c) creation of a prototype; d) user evaluation. Assets from iconfinder.com.

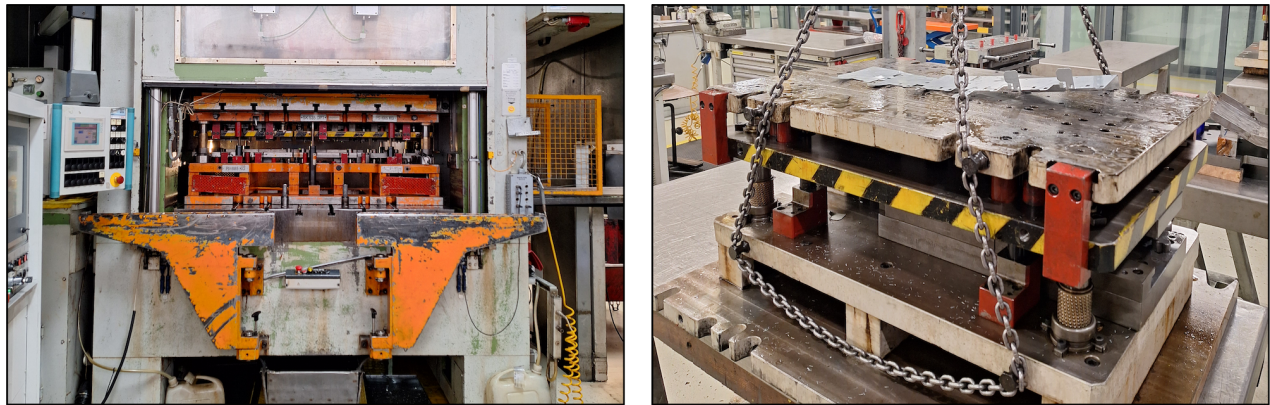


Figure 2: Use case context showcasing a heavy hydraulic press (left), crucial for the assembly line. While spare modules for cutting and shaping steel are available, maintenance in dedicated facilities (right) are essential for addressing issues and guaranteeing the factory continues producing.

designed to exert significant force to manipulate materials into precise forms. Given its critical role, maintaining its performance is essential to ensure product quality, operational efficiency, and worker safety. As such, maintenance procedures conducted by technicians are vital for preventing failures and sustaining continuous operation.

Depending on the intervention required, maintenance may sometimes involve a team of technicians performing detailed inspections across multiple subsystems, including hydraulic systems, mechanical structures, electrical wiring, and safety mechanisms. In this work, we focused on the center's piece, the module responsible for giving a specific shape to the metal sheets (see Figure 2). Either the case, effective communication is key for establishing a shared understanding of the problem at hand and determining the most suitable course of action.

To support the development of the proposed tool, various remote and in-person meetings were held with industrial partners to better understand the specific challenges and needs faced by technicians during collaborative maintenance scenarios. This effort led to the definition of key requirements that directly informed the design and implementation of the AR prototype.

Taking into consideration previous experience [4, 7, 9], it was decided that the tool would target the Microsoft HoloLens 2, a head-mounted AR-based device that allows technicians to interact with digital content overlaid onto the physical environment through spatial mapping and gesture-based controls. This hands-free interaction is particularly valuable in industrial contexts, as it enables them to operate the tool while simultaneously engaging in other activities.

The proposed solution includes a set of collaborative features

(Figure 3) accessible to all team members, allowing them to visualize and interact with shared content in a fully synchronous manner. This means that when one technician performs an action on the center's piece of the hydraulic press, all changes are reflected in real-time for the entire team. To enable this, an initial alignment is required to ensure that all participants perceive the shared content correctly positioned within the physical environment. This is achieved by scanning a QR code placed at the center of the physical space, which triggers the alignment of the shared content across all devices involved. Additionally, an ownership mechanism was implemented to maintain interaction consistency, allowing only one team member at a time to manipulate the shared content.

- **Exploded view** - The center's piece of the press can be expanded by deconstructing it into its individual components, enabling a detailed inspection of each part (Figure 3–1). This offers a clearer understanding of the structure, facilitates discussion about specific parts, and supports more precise diagnostics and planning;
- **Piece selection** - A particular part of the shared model can also be selected to facilitate closer analysis and discussion. Technicians simply point their index finger at the desired component and perform a pinch gesture, causing the selected piece to move toward them, eliminating the need for physical repositioning within the workspace (Figure 3 – 2). This ensures a more efficient interaction, allowing them to focus on critical details without disrupting the collaborative flow;

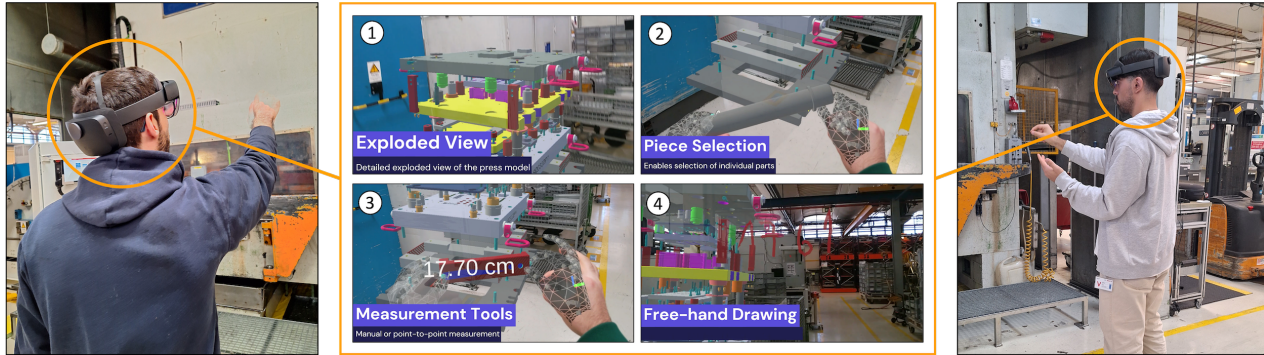


Figure 3: Overview of the co-located collaborative process, where two team members share AR content from the heavy hydraulic press module used for cutting and shaping steel. This shared view enables real-time discussion, synchronized model manipulation, and annotation.

- **Measurement Tools** - Measuring existing parts helps support maintenance planning, such as verifying whether there is enough room for the necessary tools to operate effectively (Figure 3 – 3). This enables technicians to anticipate spatial constraints during co-design, make informed decisions, and ensure that any intervention can be carried out safely and efficiently;
- **Annotations** - A key feature for expressing ideas, raising questions, and supporting the brainstorming is the ability to draw directly on the shared model using distinct colors (Figure 3 – 4). These annotations remain attached to specific parts of the model for as long as needed but can be removed at any time by the person who created them. This promotes clearer communication among team members, helps maintain context during discussions, and allows for more dynamic and personalized collaboration.

The tool was developed using the Unity game engine with the Mixed Reality Toolkit (MRTK) to enable content manipulation. To ensure real-time synchronization of shared content among team members, the Photon PUN 2 (Photon Unity Networking) was integrated. QR Code tracking was employed to align virtual content accurately within the physical environment. This was performed over Wi-Fi through a WRT 1200 AC Dual-Band Router operating on the 5GHz band.

3 ASSESSMENTS AT THE SHOP FLOOR WITH EXPERTS

The tool was tested iteratively on the shop floor with various stakeholders from the industrial sector across two separate sessions (see Figure 4). In total, 18 individuals participated in the evaluation, including technicians, the factory's maintenance supervisor, and Industry 4.0 engineers, all with prior experience relevant to the study's context. All participants had limited exposure to Augmented Reality and had engaged in collaborative activities related to the selected use case.

The primary objective of these initial assessments was to identify and address usability constraints while gathering feedback and suggestions to refine the collaborative solution under development. After providing informed consent, participants were introduced to the prototype through a short adaptation phase. They then explored its core features with a counterpart, simulating realistic maintenance and co-design activities. A researcher was present throughout to assist when necessary and to ensure the session proceeded smoothly.

Given the exploratory nature of the study and the limited availability of the participants, the focus was placed on collecting qualitative data. Observations were documented by the researcher during the sessions, and each trial concluded with a semi-structured interview

to gather detailed feedback on user experience and perceived usefulness of the tool for collaborative scenarios at the shop floor.

All teams explored the features offered by the tool. Given the participants' prior experience in co-design and maintenance, there was unanimous recognition of the tool's potential to support such scenarios effectively, noting an initial learning curve in adapting to the interaction paradigm, likely due to limited hands-on experience with AR. Nonetheless, they expressed confidence that this would be mitigated with continued use and practice.

Decomposing/manipulating complex models within the AR setting was considered more practical than traditional methods, such as using a standard computer. The spatial interaction and hands-on engagement enabled by AR allowed for a more intuitive and immersive understanding of the components. Participants also showed curiosity regarding the scalability of the system—specifically, the number of models that could be handled simultaneously and how many individuals could join the collaborative session. Due to hardware constraints, the demonstration was limited to two users at a time.

An important aspect highlighted was the ability for both participants to select and manipulate the models to express their individual viewpoints. The synchronous visual feedback offered more flexibility than sharing a single computer screen, enhancing mutual understanding. At the same time, several participants emphasized the need for a personal workspace where they could brainstorm or explore ideas independently before sharing them with their counterpart—underscoring the value of balancing shared and private modes of interaction.

The annotation feature was particularly appreciated, allowing to draw attention to specific elements, express ideas, or raise questions within the AR space. A common suggestion was to implement a different color for each user, making it easier to identify contributions and support more structured discussions, contributing to higher accountability. Additionally, some participants noted the need to adjust viewpoints for optimal sketch comprehension, suggesting that both members should be able to see the exact same perspective when one of them is creating annotations to avoid misinterpretations.

Overall, the tool was recognized as well-aligned with the requirements earlier identified earlier, being able to support collaborative activities like diagnosis, idea exchange, and problem-solving, which are essential in industrial contexts.

4 CONCLUDING REMARKS AND FUTURE WORK

Maintenance and co-design procedures involving multiple technicians are crucial in industrial settings where minimizing errors and downtime is a top priority. This work presents a shared AR tool, enabling co-located technicians to establish a shared understanding



Figure 4: Initial assessments of the AR-based shared experience for co-design and maintenance, conducted at the shop floor with domain experts.

of the task at hand. The tool supports real-time 3D object manipulation, decomposition of complex models, precise measurements, and sketching annotations, enhancing communication and coordination. Validated through iterative testing with domain experts, the tool supported real-life tasks that require team collaboration, facilitating idea exchange, and problem-solving.

Given the positive feedback, our industrial partners requested an expansion of the tool's features to support remote collaboration, enabling off-site experts to assist in problem-solving (one-to-many collaboration). We plan to have the remote expert immersed in a virtual environment with access to the same shared 3D models as the on-site technicians. Besides, images and videos and other types of annotation may be integrated to facilitate the collaborative process.

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